Advanced Applied Statistics
Lecture 5

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# Shrinkage estimation

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## Downsides of maximum-likelihood estimation

MLE estimates can be extremely off if estimated on little data.

- MLE estimate of the Bernoulli parameter of a coin flip based on a sample of one will always be exactly 0 or 1.

With large numbers of parameters, $\hat{\theta} = T(\hat{\mu})$ may be off even if each component of $\hat{\mu}$ is well estimated.

*Shrinkage estimators* trade variance for bias to address both shortcomings.

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## Bias and variance

**Bias:**
Does the expectation of the estimator match the population statistic?

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta$$

**Variance:**
What is the spread of estimates from different samples?

$$\mathrm{Var}(\hat{\theta}) = E((\hat{\theta} - E(\hat{\theta}))^2)$$

The expected estimation error can be decomposed into a *bias term*, a *variance term* and a process-inherent *irreducible error* (*aleatoric uncertainty*).

$$E((\hat{\theta} - \theta)^2) = B(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}) + \sigma^2$$

## Multivariate point estimation

- Vector of parameters $\boldsymbol{\mu} \in \mathbb{R}^N$ to be estimated from a single observation $\boldsymbol{x}$.

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_N \end{pmatrix} \qquad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix}$$

- The risk metric we care about is *expected squared error*, $E(||\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}||^2)$.
- The maximum-likelihood estimate of $\boldsymbol{\mu}$ is just $\hat{\boldsymbol{\mu}}^{\mathrm{MLE}} = \boldsymbol{x}$.
- Somewhat surprisingly, in multiple dimensions we can do better!



## James-Stein estimator

$$\hat{\boldsymbol{\mu}}^{\mathrm{JS}} = \bar{\boldsymbol{x}} + \left(1 - \frac{N-3}{||\boldsymbol{x} - \bar{\boldsymbol{x}}||^2}\right)(\boldsymbol{x} - \bar{\boldsymbol{x}})$$

- For $N \geq 4$, the James-Stein estimate has **lower squared error risk** than the ML estimate, *no matter what the value of* $\boldsymbol{x}$ *is.*

$$E_\mu(||\hat{\boldsymbol{\mu}}^{\mathrm{JS}} - \boldsymbol{\mu}||^2) < E_\mu(||\hat{\boldsymbol{\mu}}^{\mathrm{MLE}} - \boldsymbol{\mu}||^2)$$

- The ML estimate $\hat{\boldsymbol{\mu}} = \boldsymbol{x}$ undergoes **shrinkage** by a factor of

$$1 - \frac{N-3}{||\boldsymbol{x} - \bar{\boldsymbol{x}}||^2}.$$

- Learning from other instances:
  - James-Stein estimation is only possible in a multidimensional setting.
  - Only consistently better than MLE for $N \geq 4$.

## Bayesian derivation

- Assume hierarchical normal model:

$$\mu_i \sim N(M, A) \qquad x_i|\mu_i \sim N(\mu_i, 1)$$

- Then the Bayesian estimate is

$$\hat{\boldsymbol{\mu}}^{\text{Bayes}} = \boldsymbol{M} + B(\boldsymbol{x} - \boldsymbol{M}), \text{ with } B = \frac{A}{A+1}.$$

- Marginalising out $\mu_i$, we obtain

$$x_i \sim N(M, A+1).$$



## Bayesian derivation

$$x_i \sim N(M, A+1).$$

- $\hat{M} = \bar{x}$ is an unbiased estimate of $M$.
- Less obviously, the following $\hat{B}$ is an unbiased estimate of $B$:

$$\hat{B} = 1 - \frac{N-3}{\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

This follows because $S = \sum_i (x_i - \bar{x})^2 \sim (A+1)\chi^2_{N-1}$ and

$$B = \frac{A}{A+1} = 1 - \frac{1}{A+1} \quad \text{and} \quad (N-3)E\left(\frac{1}{S}\right) = \frac{1}{A+1}.$$

- Plugging this into the Bayes estimate yields the James-Stein estimator.

$$\hat{\boldsymbol{\mu}}^{\text{JS}} = \hat{\boldsymbol{M}} + \hat{B}(\boldsymbol{x} - \hat{\boldsymbol{M}})$$

## Trading off bias and variance

- Bayesian derivation assumed clustered means, $\mu_i \sim N(M, A)$.
- But the James-Stein theorem is valid *generally*,
  even if means aren't well-clustered.

Assume general shrinkage estimator $\hat{\boldsymbol{\theta}}_\lambda = \lambda \boldsymbol{X}$ and $X_i \sim N(\theta_i, 1)$.

- Mean squared error can be decomposed into *bias* and *variance* terms.

$$\begin{aligned}\sum_{i=1}^{N} E\left((\hat{\theta}_{\lambda,i} - \theta_i)^2\right) &= \sum_{i=1}^{N}\left(E(\hat{\theta}_i) - \theta_i\right)^2 + \sum_{i=1}^{N} E\left((\hat{\theta}_i - E(\hat{\theta}_i))^2\right) \\ &= \underbrace{(\lambda-1)^2||\boldsymbol{\theta}||^2}_{\text{Bias}^2} + \underbrace{N\lambda^2}_{\text{Variance}}\end{aligned}$$

- Bias depends only on (true) $\boldsymbol{\theta}$, variance depends only on $N$.

https://www.jchau.org/2021/01/29/demystifying-stein-s-paradox/

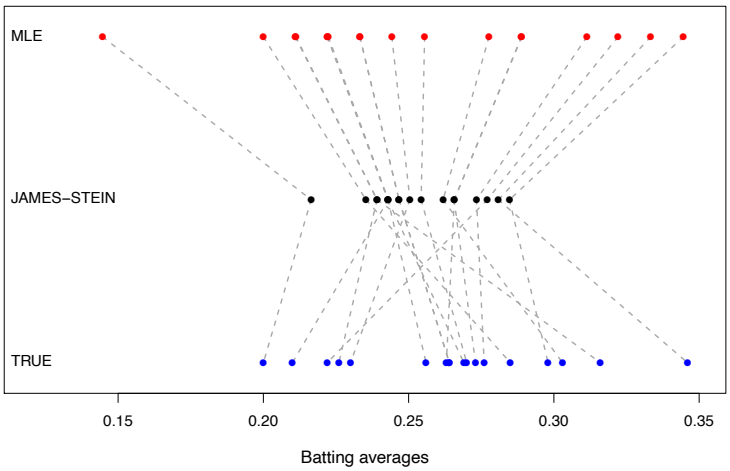



**Figure 7.1** Eighteen baseball players; top line MLE, middle James–Stein, bottom true values. Only 13 points are visible, since there are ties.



## Application: Finding associations between genes

- The *entropy* of a random variable quantifies the average amount of information associated with its possible outcomes.

$$H(X) = -\sum_{x \in \mathcal{X}} \theta(x) \log \theta(x)$$

- *Mutual information* measures the mutual dependence between two variables.

$$I(X, Y) = H(X) + H(Y) - H(X, Y)$$

- How can we get good estimates of the entropy to identify associations between outcomes?

Hausser and Strimmer, ArXiv 0811.3579



## Application: Estimating entropy from small samples

- Scenario: Discrete space with $p$ possible outcomes, each attested $y_k$ times ($k = 1, \ldots, p$).
- Total number of observations:

$$n = \sum_{k=1}^{p} y_k$$

- MLE of $\hat{\theta}_k$ is

$$\hat{\theta}_k^{\mathsf{ML}} = \frac{y_k}{n}.$$

- Plug-in estimate of entropy is

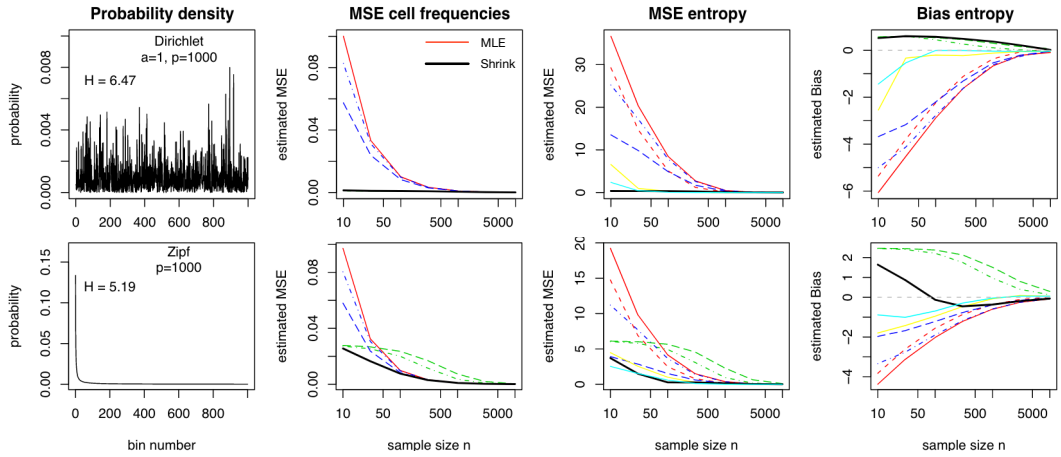
$$\hat{H}^{\mathsf{ML}} = -\sum_{k=1}^{p} \hat{\theta}_k^{\mathsf{ML}} \log \hat{\theta}_k^{\mathsf{ML}}.$$

# Application: Estimating entropy from small samples

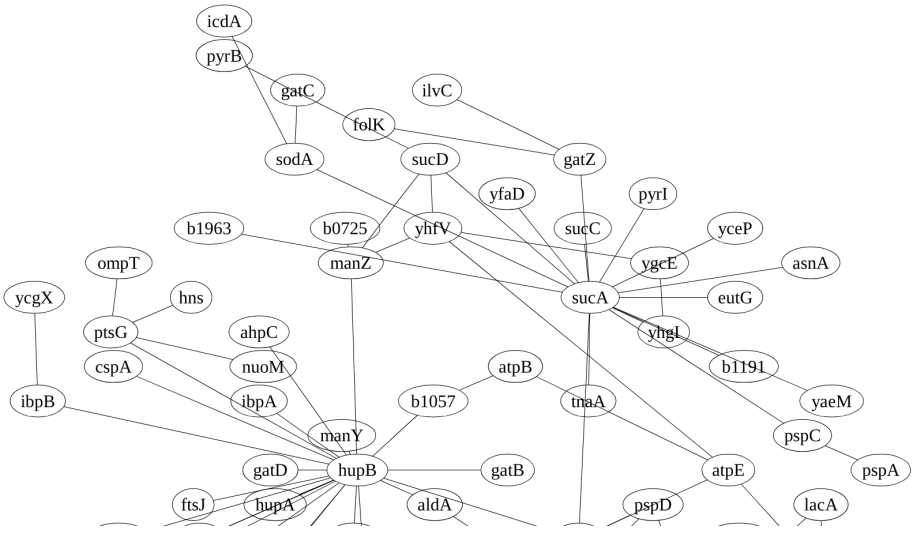
- Plug-in estimator of entropy is good if *all cells are well represented in the data*.
- If $n \ll p$ (many zero counts), it **seriously underestimates the entropy**.
- Hausser and Strimmer derive a James-Stein estimator of the probability that interpolates between a uniform *shrinkage target* and the MLE.

$$\hat{\theta}_k^{\text{Shrink}} = \frac{\lambda}{p} + (1-\lambda)\hat{\theta}_{\text{ML}} = \frac{1}{p} + \lambda' \left( \hat{\theta}_{\text{ML}} - \frac{1}{p} \right).$$

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Hausser and Strimmer, ArXiv 0811.3579




Hausser and Strimmer, ArXiv 0811.3579

# Regularisation



## Limiting model capacity

- We want our models to learn generalisable patterns, not idiosyncrasies of samples.
- Key idea of regularisation: Limit the capacity of the model, so it only learns patterns for which it has good evidence.
- Artificially introduce **bias** to reduce the variance.
  - Bias: Does the expectation of the estimator match the population statistic?

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta$$

  - Variance: What's the spread of estimated from different samples?

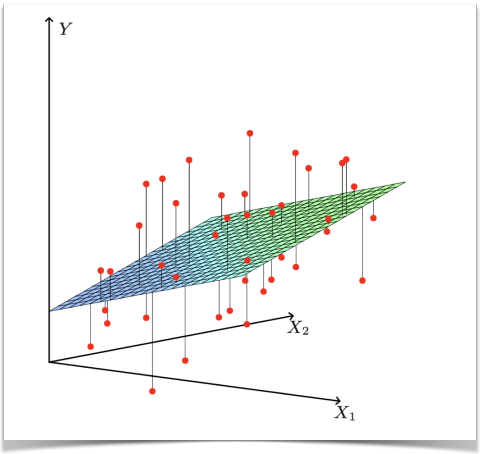


## Example: Linear regression

$$f(\mathbf{x}; \beta) = \beta_0 + \sum_i x_i \beta_i$$

Usually trained by minimising the **squared loss** or **residual sum of squares**:

$$\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{N} (y_i - f(\mathbf{x}_i; \beta))^2$$



## Vector norms

The norm of a vector is a way of measuring its length.
The most important norms can be written as

$$||x||_p = \left( \sum_{i=1}^{n} |x_i|^p \right)^{\frac{1}{p}}$$

**$\ell_1$ norm**

$$||x||_1 = \sum_{k=1}^{d} |x_k|$$

**$\ell_2$ norm**

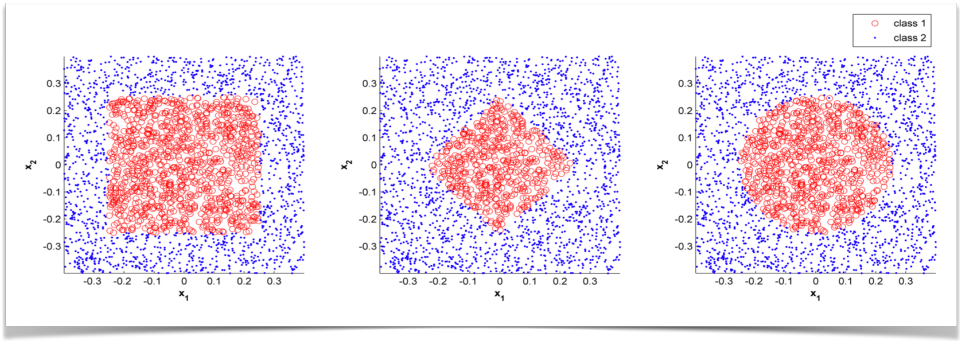
$$||x||_1 = \sqrt{\sum_{k=1}^{d} |x_k|^2}$$

**$\ell_\infty$-norm**

$$||x||_\infty = \max(|x_k|, \ldots, |x_k|)$$



## The "unit ball" in different norms




$||x||_\infty = \max(|x_k|, \ldots, |x_k|)$ $||x||_1 = \sum_{k=1}^{d} |x_k|$ $||x||_2 = \sqrt{\sum_{k=1}^{d} |x_k|^2}$



## Ridge regression

Ridge regression is linear regression with squared loss
and a penalty on the $\ell_2$-norm of $\beta$.

$$\beta_{\text{ridge}} = \arg\min_{\beta} \left( \sum_{i=1}^{N} (y_i - f(\mathbf{x}_i))^2 + \lambda ||\beta||_2^2 \right)$$

Effects of ridge regression:

- Biased parameter estimates:
  Estimated coefficients are shrunk towards zero.
- The model requires "stronger evidence" to grow the weights.
- Avoids overfitting.
- Potentially better performance on new datasets.

## Ridge regression as constrained optimisation

- Ridge regression is equivalent to a constrained optimisation problem:
  Minimise $\sum_{i=1}^{N}(y_i - f(\mathbf{x}_i))^2$ subject to $||\beta||_2^2 \leq c$.
- $\lambda$ is just a Lagrange multiplier!
- Instead of fixing the constraint $c$ and optimising $\lambda$,
  we usually fix $\lambda$, and the constraint is variable.



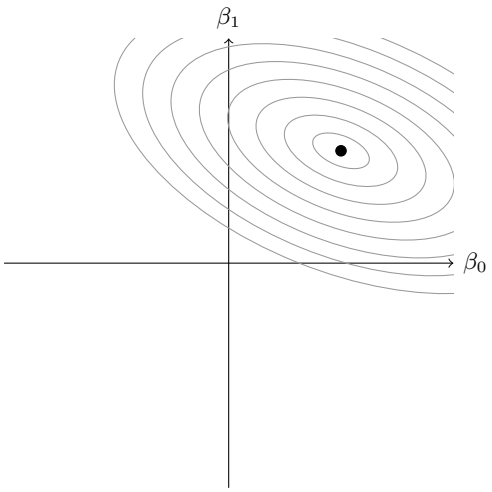
## LASSO regression

$$\beta_{\text{LASSO}} = \arg\min_{\beta} \left( \sum_{i=1}^{N} (y_i - f(\mathbf{x}_i))^2 + \lambda ||\beta||_1 \right)$$

- LASSO regression uses squared loss and a penalty on the $\ell_1$ norm of $\beta$.
- $\ell_1$ regularisation induces **sparse** solutions:
  Many parameters will be 0.
- Built-in feature selection!



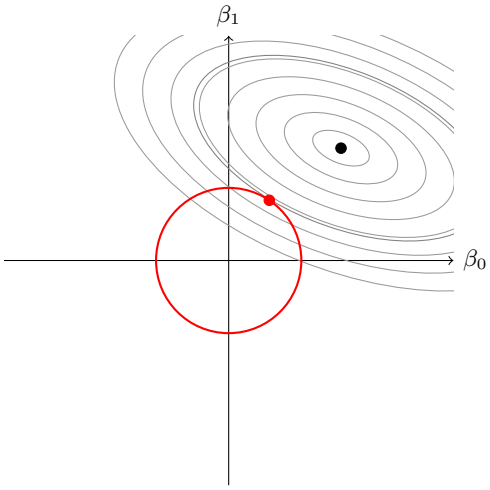
## Geometric view of regularisation




Without regularisation, the square loss defines a bowl-shaped surface.

In this 2D example, its contour lines are ellipses.

## Geometric view of regularisation



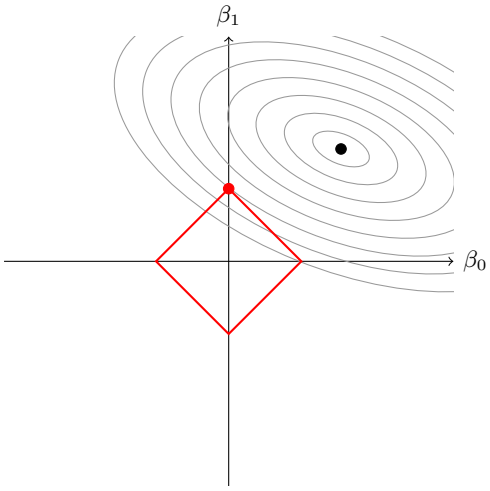

Ridge regression ($\ell_2$ regularisation) constrains the weights to be in a ball centered at the origin.

The solution is the closest point from the unconstrained solution that satisfies the constraint.

The weights will be smaller, but generally non-zero.



## Geometric view of regularisation




LASSO regression ($\ell_1$ regularisation) imposes a constraint with sharp corners.

There's a high probability that many parameters will be exactly zero.

This produces a sparse solution.